- Used in solid state video displays and optical communication system.
- Used in programmable advertisement boards.
- Used in image sensing circuit.

10.7 LIQUID CRYSTAL DISPLAY(LCD)

Liquid crystal display (LCDs) are used for display of numberic and alphanumeric character in dot matrix and segmental displays. The two liquid crystal materials which are commonly used in display technology are nematic and cholesteric.

Advantages of LCD

1) The voltage required are small.
2) They have a low power consumption. A seven segment display requires about 140W (20W per segment), whereas LCDs require about 40mw per numeral.
3) They are economical.

Disadvantages of LCD

1) LCDs are very slow device. The Turn 'on' and 'off' times are quite large.
2) When used in dc, their life span is quite small. Therefore, they are used with ac supplies having a frequency less than 50 HZ.
3) They occupy a large area.

10.8 SEVEN SEGMENT DISPLAY

One way of producing an alphanumeric display is to make a seven segment monolithic device as shown in fig 10.12 which can display all numerals. Each segment contains LED which can be turned 'on' or 'off' to form the desired digit. Each segment of the array has to be switched in response to a logic signal.

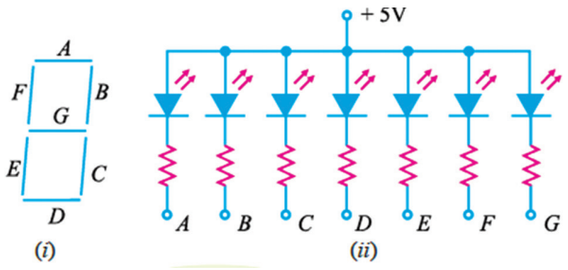


Fig 10.12 *7-Segment LED Display Common Anode*

For example Fig 10.12 (i) shows the response to a logic signal corresponding to 2, in which segments a ,b, g, e and d have been switched 'on' and c and f remains 'off'. Similarly when all segments are 'on', the digit formed is 'off' 8. If only the center segment g is the digit will be zero. Common anode and common cathode seven segment LED displays are shown in Fig 10.12 (ii) common anode type LED displays an active 'low' configuration, where as an active 'high' circuitry is necessary for the common cathode type LED display.

10.9 CCTV CAMERAS

CCTV systems have become extremely popular over a decade as the technology has improved and become more affordable. Themajority of CCTV camera in use nowdays are usually for surveillance and security purposes. CCTV systems can be found in almost every bank, casino, mall and large departmental store. In fact,

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